ESM-2 single-point mutation scan

WP_018752751.1 Paenibacillus sanguinis · 318 aa · generated 2026-11-10 17:09

1. What was run

A masked-marginal scan over every residue of the query. For each position the residue is replaced by `<mask>`, the full-length sequence is passed through ESM-2, and the softmax over the 20 amino acids at that position gives a distribution. The scoring follows `mutation_prediction.py` in `eust-w/esm2agent` so the numbers stay comparable with the earlier run in that repository.

Model	esm2_t33_650M_UR50D
Parameters	651M (651,043,254)
Checkpoint	huggingface.co/facebook/esm2_t33_650M_UR50D (model.safetensors), the transformers conversion of the fair-esm checkpoint
Weight SHA-256	`a08adabb949fa67ad3c14b509d04fd60368b35007b0095e3358f81200c4f4db0`
Method	masked-marginal, one forward pass per position, full-length context
Forward passes	318 (one per residue), batch size 4
Candidates kept	top 5 non-wild-type residues per position = 1590 total
Device	cpu on AMD64 Family 23 Model 113 Stepping 0, AuthenticAMD
Runtime	628.6 s
Reproduced from	github.com/eust-w/esm2agent · mutation_prediction.py

2. Two rankings, and which one to use

By p(mut) is the repository's original ranking: the raw masked probability of the mutant residue. It answers "where does the model most confidently expect some other residue?", so it is dominated by positions the model thinks are mis-assigned in the wild type, and it ignores how likely the wild-type residue was.

By LLR is log p(mut) – log p(wt), the standard variant-effect score. It measures how much the model prefers the mutant *over the residue actually there*, which is the quantity that correlates with measured fitness in the ESM papers and in ProteinGym. Both tables are included; LLR is the one to act on.

Caveat worth stating plainly: a high LLR means the substitution looks natural to a language model trained on UniRef. That is evidence about evolutionary plausibility, not about activity, stability, or expression. Catalytic and binding residues are often highly conserved, so the model will correctly refuse to mutate them — but it will also happily propose substitutions that are plausible and functionally useless. These 100 are a shortlist to test, not a set of predictions.

3. Query sequence

```
1 MANDRIEELL KYQGSSPKPE DFDAYWERAL KELEAQSLDY ELVAADFQTP MAECYHLYFT
61 GVGGARVHAK YVKPRAAQGK GPGLALFHGY HSGSGDWSDK AAYAAHGITV LAMDCRGQGG
121 LSEDHLTVKG TTLKGHIIRG VDEDNPDKLY YRNVFLDLVQ TVRILMSMEQ VDPERIGAHG
181 CSQGGALTIA CAALEPRVAL AIPVYPFLSD YQKAWEMDVT ISAYEELAYY FRFMDPLHRT
241 QEAFFNKLGY IDVQNLADRI RAKVVFVTGL ADTVCPPYTQ FAAYNKMASD KELVVYHEYG
301 HEYLPFLSDR VLEEMLKL
```

4. Sanity checks run before trusting the ranking

H297P came out first by a wide margin, at a position that superficially looks catalytic. That is the signature of a mis-loaded model or an off-by-one in the token offset, so the run was checked before the tables were accepted.

Token offset	318 positions checked; mismatches: none
Unmasked identity	0.962 (the model reproduces the sequence it is shown)
Masked wild-type recovery	0.575 (ESM-2 650M on a natural protein: 0.45-0.65)

S182 nucleophile	p(wt) = 0.9990 — conserved, as a real catalytic serine must be
H301 catalytic His	p(wt) = 0.9997 — conserved
H297	p(wt) = 0.0014, model prefers Pro at 0.946 — not conserved by this model

Verdict. The weights load correctly and the model discriminates: it refuses to substitute the two residues a Ser-Asp-His hydrolase needs, while freely substituting H297. So H297P ranking first is a genuine preference of the model, not a loading artefact. It is still only a statement about sequence plausibility under a UniRef-trained language model.

5. What the conservation pattern says about this protein

GxSxG nucleophile elbow	`GCSQGG` at 180-185. The alpha/beta hydrolase nucleophile elbow; S182 is the catalytic serine. The scan holds S182 at p(wt)=0.999, so the model agrees it is not substitutable.
Catalytic histidine	H301, inside `HEYGHE` at 297-302, is held at p(wt)=0.9997. With S182 this is consistent with a Ser-Asp-His triad.
HExxH resemblance is coincidental	`HEYGHE` superficially matches the HExxH metallopeptidase signature, which would make H297 a metal-coordinating residue. The scan contradicts that: H297 has p(wt)=0.0014 and the model prefers Pro at 0.946, while the residues a hydrolase triad needs (S182, H301) are the conserved ones. Read as an alpha/beta hydrolase, not a metallopeptidase. Sequence-level inference only; no structure was used.

Per-position substitution landscape

LLR for all 20 amino acids at every position. Blue = the model prefers the substitution over the wild type; red = it prefers the wild type. Dark vertical bands are conserved positions.



Top 100 by LLR = log p(mut) − log p(wt)

The recommended ranking. Ranks 1–50 on the left, 51–100 on the right.

#	Mutation	LLR	p(mut)	p(wt)
1	H297P	+6.499	0.9461	0.0014
2	L158A	+5.705	0.7449	0.0025
3	M287I	+5.676	0.8734	0.0030
4	M234R	+5.112	0.5346	0.0032
5	C181G	+4.967	0.7603	0.0053
6	D4L	+4.751	0.3784	0.0033
7	M234H	+4.668	0.3428	0.0032
8	H91G	+4.640	0.5400	0.0052
9	D4M	+4.523	0.3014	0.0033
10	G81L	+4.087	0.4229	0.0071
11	L158C	+4.036	0.1405	0.0025
12	F306G	+4.030	0.8218	0.0146
13	T161A	+3.866	0.6124	0.0128
14	I108Y	+3.833	0.5621	0.0122
15	T240R	+3.653	0.2182	0.0057
16	Q241E	+3.590	0.9101	0.0251
17	H91T	+3.579	0.1870	0.0052
18	T240E	+3.554	0.1977	0.0057
19	M166A	+3.490	0.2591	0.0079
20	Y303D	+3.441	0.4209	0.0135
21	K129P	+3.418	0.6301	0.0207
22	T240Q	+3.412	0.1715	0.0057
23	D4Y	+3.272	0.0862	0.0033
24	L200F	+3.223	0.7094	0.0283
25	S37P	+3.202	0.7985	0.0325
26	M287V	+3.173	0.0715	0.0030
27	M166C	+3.172	0.1886	0.0079
28	Y278S	+3.096	0.3263	0.0148
29	G81M	+3.059	0.1512	0.0071
30	L158G	+3.008	0.0502	0.0025
31	I108F	+2.996	0.2433	0.0122
32	V219L	+2.979	0.5931	0.0302
33	H106A	+2.923	0.4191	0.0225
34	T240K	+2.922	0.1051	0.0057
35	Y303E	+2.887	0.2418	0.0135
36	S289C	+2.867	0.8076	0.0459
37	M166Q	+2.833	0.1344	0.0079
38	A178V	+2.813	0.8011	0.0481
39	C191A	+2.812	0.4373	0.0263
40	M287L	+2.795	0.0490	0.0030
41	T240N	+2.782	0.0914	0.0057
42	Y278E	+2.749	0.2307	0.0148
43	A199K	+2.674	0.7655	0.0528
44	H91S	+2.656	0.0743	0.0052
45	A101L	+2.652	0.6633	0.0468
46	H179T	+2.610	0.2041	0.0150
47	T161G	+2.609	0.1742	0.0128
48	D4I	+2.579	0.0431	0.0033
49	T161S	+2.574	0.1683	0.0128
50	A44P	+2.561	0.7259	0.0561

#	Mutation	LLR	p(mut)	p(wt)
51	Y71L	+2.555	0.8242	0.0640
52	A271M	+2.532	0.5614	0.0446
53	V265L	+2.432	0.7267	0.0639
54	Q170E	+2.356	0.5282	0.0501
55	M166R	+2.329	0.0812	0.0079
56	C191L	+2.327	0.2691	0.0263
57	H125P	+2.317	0.1354	0.0133
58	G81T	+2.284	0.0697	0.0071
59	I108W	+2.271	0.1179	0.0122
60	M166S	+2.234	0.0738	0.0079
61	V67I	+2.225	0.8799	0.0950
62	N3F	+2.216	0.2694	0.0294
63	H91A	+2.208	0.0475	0.0052
64	G81A	+2.203	0.0643	0.0071
65	H125A	+2.183	0.1184	0.0133
66	D4F	+2.169	0.0286	0.0033
67	M315V	+2.167	0.2675	0.0306
68	C181S	+2.160	0.0459	0.0053
69	L158T	+2.150	0.0213	0.0025
70	Y278A	+2.149	0.1267	0.0148
71	H125E	+2.146	0.1141	0.0133
72	H56D	+2.141	0.8308	0.0976
73	H297E	+2.136	0.0121	0.0014
74	C181A	+2.129	0.0445	0.0053
75	V198I	+2.119	0.8296	0.0997
76	I221G	+2.112	0.1156	0.0140
77	H179V	+2.108	0.1235	0.0150
78	I221H	+2.074	0.1112	0.0140
79	S92G	+2.062	0.6246	0.0794
80	Q36V	+2.060	0.3842	0.0490
81	M315L	+2.042	0.2360	0.0306
82	H179A	+2.035	0.1148	0.0150
83	R5P	+2.023	0.4111	0.0543
84	L38P	+2.019	0.5562	0.0739
85	A85V	+2.016	0.5933	0.0790
86	L158V	+1.987	0.0181	0.0025
87	N246R	+1.986	0.2561	0.0352
88	G81V	+1.983	0.0516	0.0071
89	I221D	+1.978	0.1011	0.0140
90	H125T	+1.977	0.0964	0.0133
91	M315I	+1.961	0.2175	0.0306
92	M234K	+1.951	0.0226	0.0032
93	I221N	+1.908	0.0942	0.0140
94	M51V	+1.905	0.2794	0.0416
95	H125D	+1.896	0.0889	0.0133
96	M168L	+1.893	0.5296	0.0798
97	K148T	+1.882	0.2618	0.0399
98	H91N	+1.881	0.0342	0.0052
99	I164L	+1.837	0.4381	0.0698
100	M234Y	+1.834	0.0201	0.0032

Full table of all 1590 candidates: `esm2_all_mutations.csv`

Top 100 by p(mut) — the repository's original ranking

Included for continuity with the earlier esm2agent run. See section 2 for why this ranking is the weaker of the two.

#	Mutation	LLR	p(mut)	p(wt)
1	H297P	+6.499	0.9461	0.0014
2	Q241E	+3.590	0.9101	0.0251
3	V67I	+2.225	0.8799	0.0950
4	M287I	+5.676	0.8734	0.0030
5	H56D	+2.141	0.8308	0.0976
6	V198I	+2.119	0.8296	0.0997
7	Y71L	+2.555	0.8242	0.0640
8	F306G	+4.030	0.8218	0.0146
9	S289C	+2.867	0.8076	0.0459
10	A178V	+2.813	0.8011	0.0481
11	S37P	+3.202	0.7985	0.0325
12	A186G	+1.387	0.7941	0.1984
13	Y25F	+1.414	0.7906	0.1923
14	A199K	+2.674	0.7655	0.0528
15	C181G	+4.967	0.7603	0.0053
16	L158A	+5.705	0.7449	0.0025
17	V265L	+2.432	0.7267	0.0639
18	A44P	+2.561	0.7259	0.0561
19	D147E	+1.181	0.7223	0.2218
20	L200F	+3.223	0.7094	0.0283
21	A101L	+2.652	0.6633	0.0468
22	E53D	+0.883	0.6626	0.2740
23	K129P	+3.418	0.6301	0.0207
24	I6L	+1.548	0.6300	0.1340
25	S92G	+2.062	0.6246	0.0794
26	E195D	+0.608	0.6176	0.3361
27	T161A	+3.866	0.6124	0.0128
28	A85V	+2.016	0.5933	0.0790
29	V219L	+2.979	0.5931	0.0302
30	K213R	+1.630	0.5646	0.1106
31	I108Y	+3.833	0.5621	0.0122
32	A271M	+2.532	0.5614	0.0446
33	L38P	+2.019	0.5562	0.0739
34	G83A	+0.356	0.5544	0.3884
35	H91G	+4.640	0.5400	0.0052
36	M234R	+5.112	0.5346	0.0032
37	S209C	+0.379	0.5313	0.3637
38	M168L	+1.893	0.5296	0.0798
39	Q170E	+2.356	0.5282	0.0501
40	E169P	+0.652	0.5268	0.2745
41	R75K	+1.710	0.5176	0.0936
42	Y55F	+0.504	0.5130	0.3099
43	Q212R	+1.454	0.5096	0.1190
44	V141F	+1.729	0.5005	0.0888
45	Y299F	+0.208	0.4900	0.3979
46	F155Y	+0.164	0.4870	0.4132
47	M217L	+0.461	0.4706	0.2968
48	L256F	-0.009	0.4648	0.4691
49	R261K	+0.572	0.4605	0.2599
50	V294I	+1.412	0.4548	0.1109

#	Mutation	LLR	p(mut)	p(wt)
51	G249S	+0.153	0.4535	0.3891
52	C115V	+0.261	0.4397	0.3387
53	I164L	+1.837	0.4381	0.0698
54	C191A	+2.812	0.4373	0.0263
55	F47Y	-0.023	0.4368	0.4471
56	D142E	+0.459	0.4293	0.2714
57	G81L	+4.087	0.4229	0.0071
58	Y303D	+3.441	0.4209	0.0135
59	H106A	+2.923	0.4191	0.0225
60	M113V	+0.803	0.4178	0.1872
61	K263R	-0.247	0.4146	0.5307
62	R5P	+2.023	0.4111	0.0543
63	V72I	+0.457	0.4053	0.2567
64	Q13K	+1.069	0.3902	0.1340
65	A223S	+0.180	0.3883	0.3242
66	Q36V	+2.060	0.3842	0.0490
67	D309E	+0.660	0.3808	0.1968
68	A105S	-0.036	0.3805	0.3943
69	L227I	-0.315	0.3791	0.5196
70	D4L	+4.751	0.3784	0.0033
71	K31A	+1.387	0.3771	0.0942
72	I164V	+1.685	0.3764	0.0698
73	D290E	+1.088	0.3740	0.1259
74	F59Y	-0.400	0.3731	0.5566
75	K247T	+0.617	0.3721	0.2008
76	E20A	+0.130	0.3715	0.3262
77	Q254M	-0.085	0.3638	0.3963
78	S94G	+0.599	0.3625	0.1992
79	A214V	+0.352	0.3614	0.2543
80	L10K	+0.435	0.3610	0.2336
81	A65S	+0.398	0.3521	0.2365
82	E143D	+0.113	0.3500	0.3127
83	K70R	+0.186	0.3482	0.2891
84	L304I	+0.001	0.3478	0.3476
85	G177A	+0.542	0.3454	0.2008
86	A29T	+0.070	0.3446	0.3215
87	A52V	+0.958	0.3429	0.1315
88	M234H	+4.668	0.3428	0.0032
89	A243E	+0.485	0.3376	0.2079
90	D99E	+0.557	0.3359	0.1924
91	V62F	+1.602	0.3341	0.0673
92	N145D	+1.573	0.3292	0.0683
93	Y278S	+3.096	0.3263	0.0148
94	E242D	-0.506	0.3258	0.5403
95	K73R	+0.517	0.3256	0.1942
96	V62Y	+1.565	0.3221	0.0673
97	E298D	-0.196	0.3152	0.3834
98	A201C	-0.622	0.3108	0.5789
99	D46P	-0.459	0.3084	0.4882
100	N153R	+0.488	0.3066	0.1881